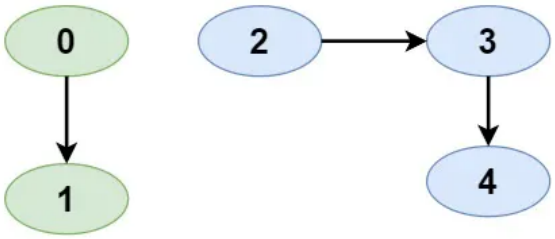
Number of Connected Components in an Undirected Graph

Problem Statement

Given an undirected graph represented by 'n' nodes labeled from `0` to `n-1` and a list of undirected edges (each edge is a pair of nodes), determine the number of connected components in the graph. A connected component is a group of nodes that are directly or indirectly linked to each other through the edges.

Examples

1. **Example 1**
   - **Input**: `n = 5`, `edges = [[0,1], [2,3], [3,4]]`



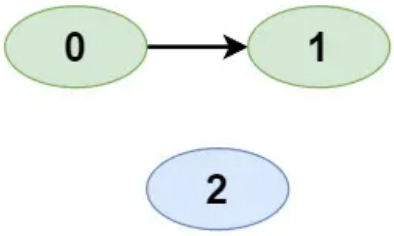
- **Expected Output**: `2`
- **Justification**: Two components are: `0-1`, and `2-3-4`.

2. **Example 2**
   - **Input**: `n = 4`, `edges = [[0,1], [1,2], [2,3]]`



- **Expected Output**: `1`
- **Justification**: All the nodes are connected in a single chain: `0-1-2-3`.

3. **Example 3**
    - **Input**: `n = 3`, `edges = [[0,1]]`



- **Expected Output**: `2`
- **Justification**: Two connected components exist: `0-1` and an isolated node `2`.

Try it yourself

Try solving this question here:

Java

```
public class Solution {
    public int countComponents(int n, int[][] edges) {
        // ToDo: Write Your Code Here.
        return n;
    }
}
```

Test Case | Run Code Result | Stdout

```
4
[[0,1],[2,3]]
```

Use Example Testcases | Run | Submit